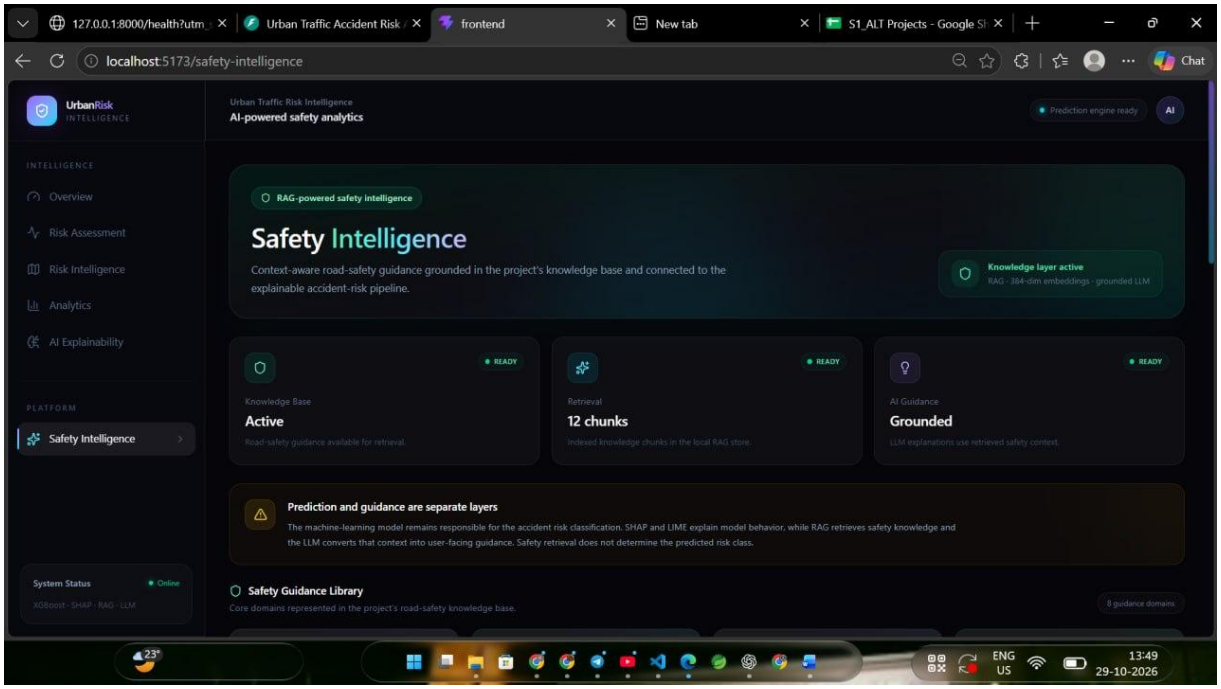
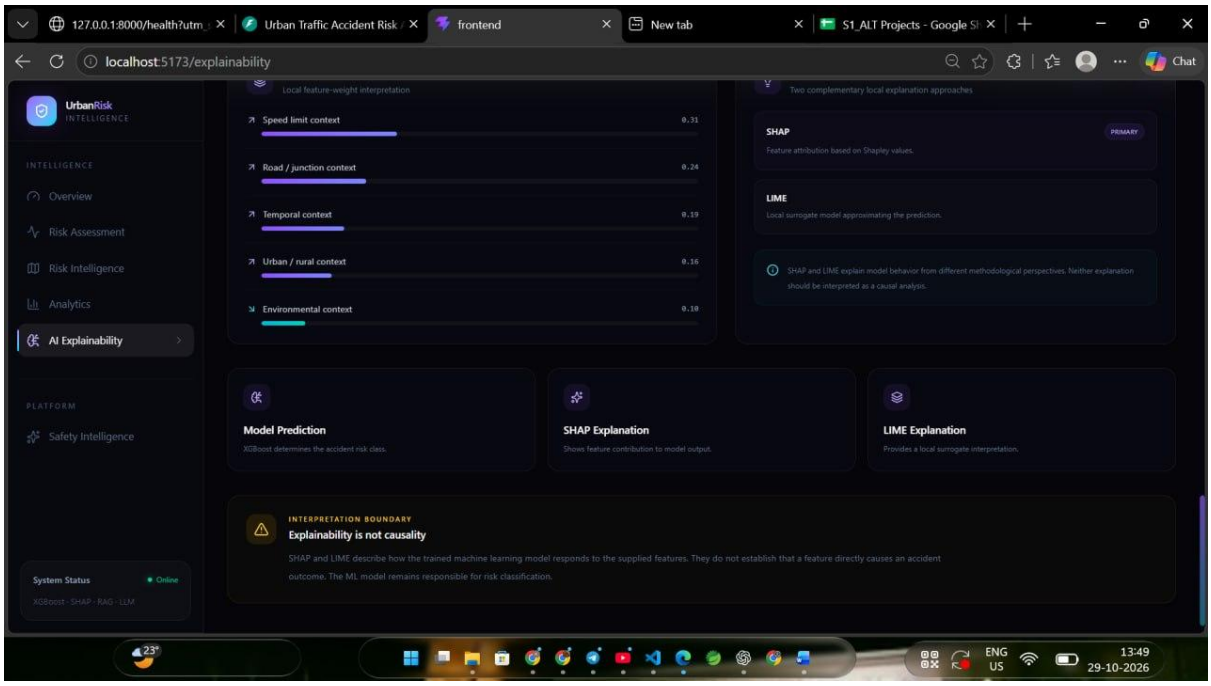
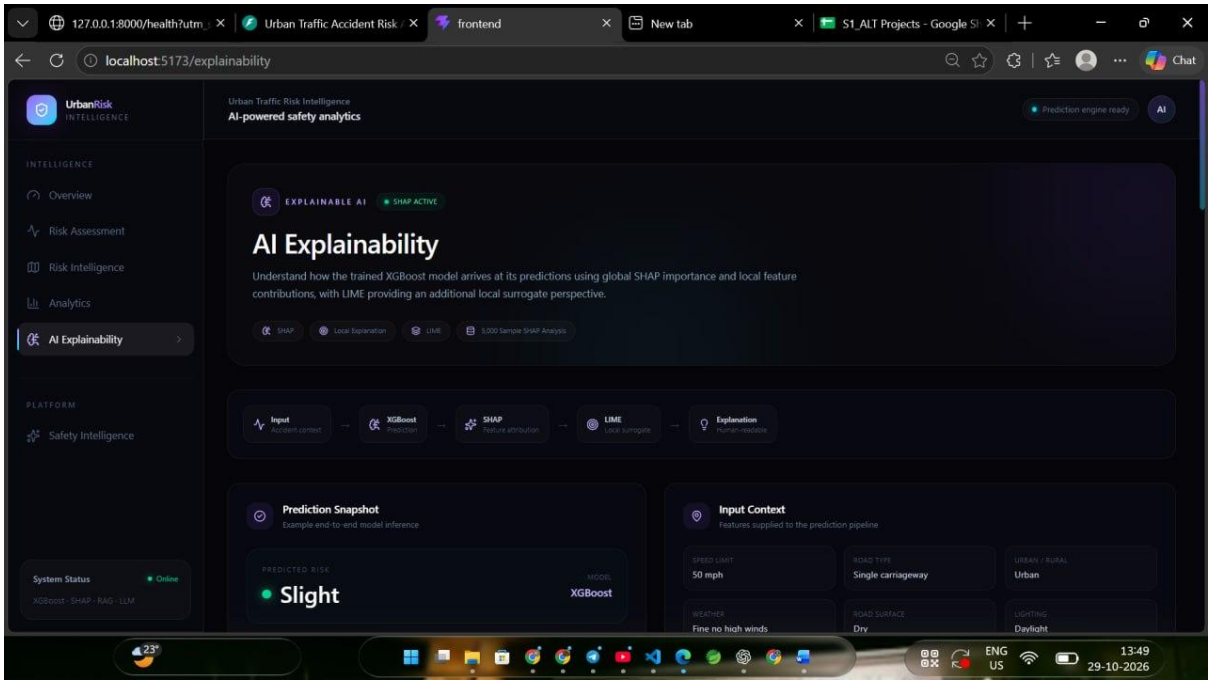
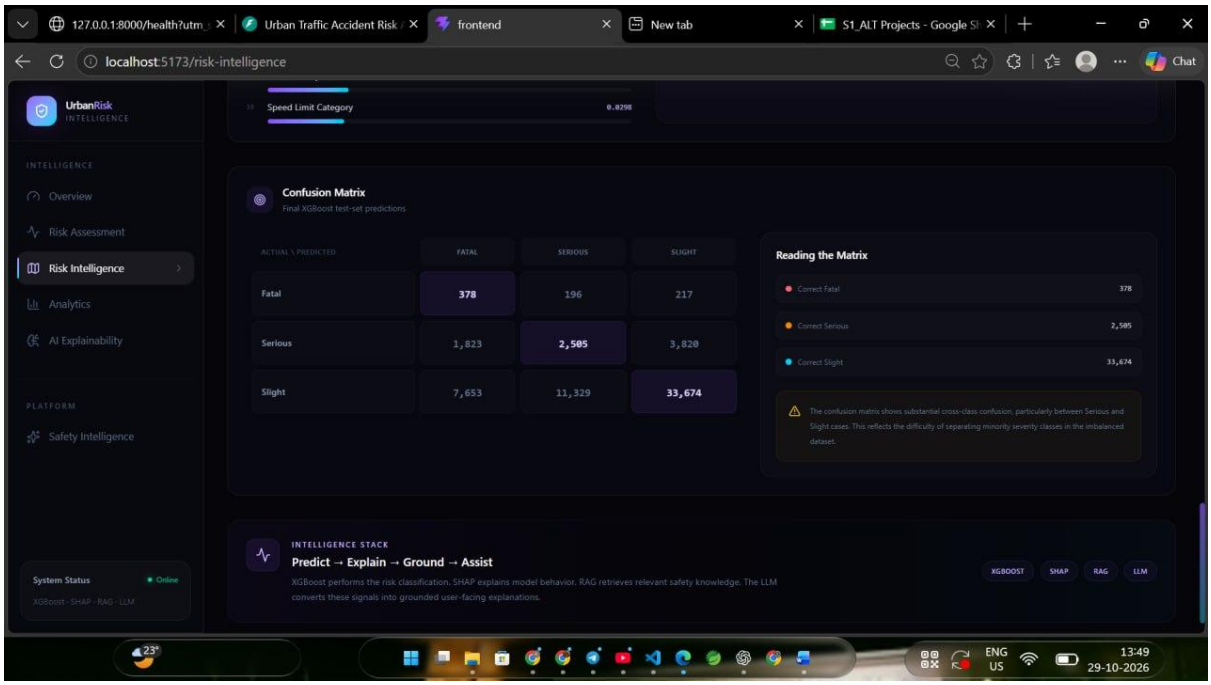
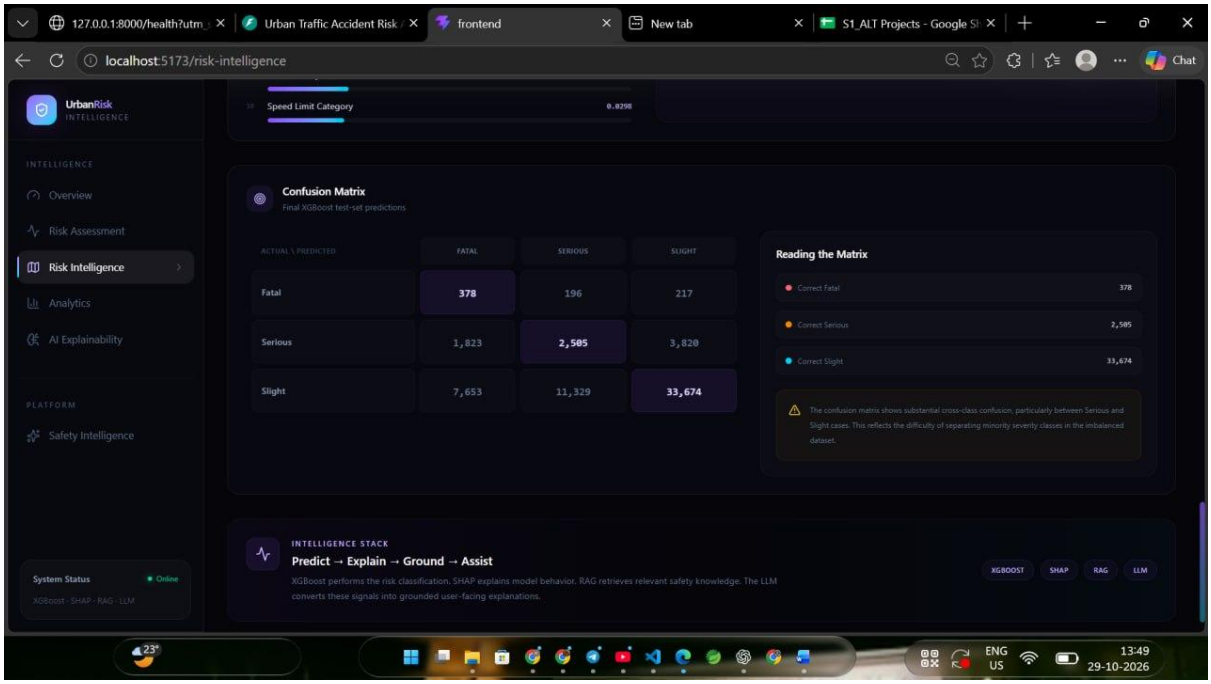
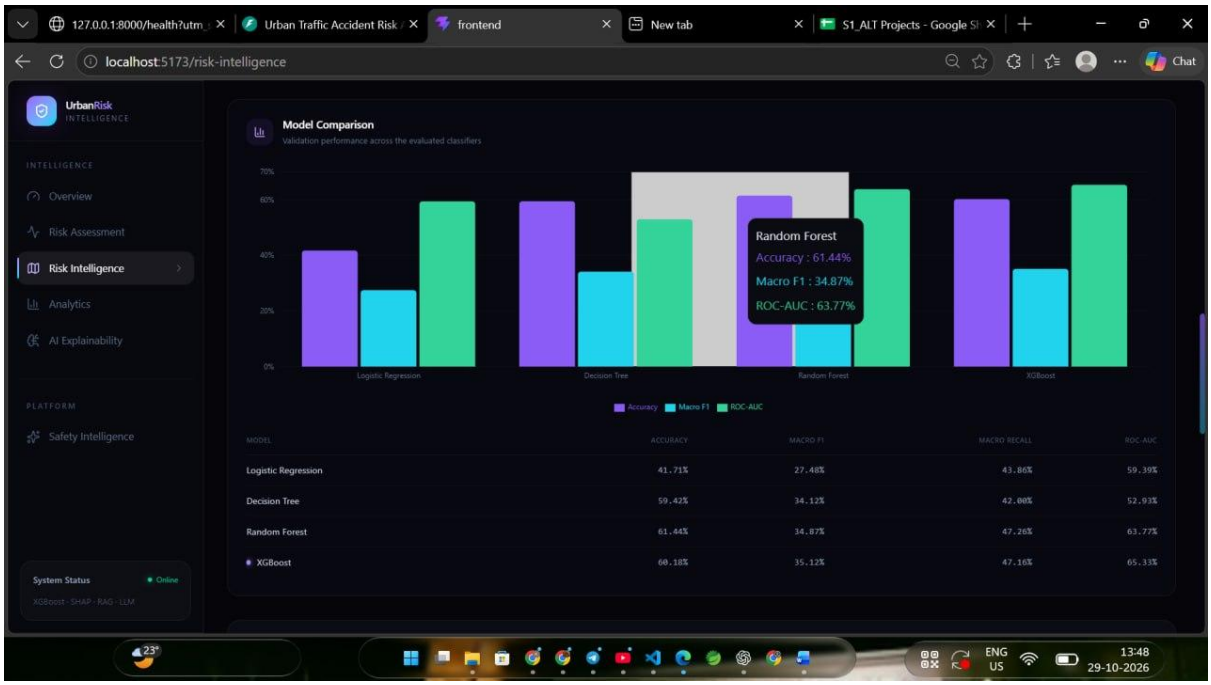

127.0.0.1:8000/health?utm
Urban Traffic Accident Risk
frontend
New tab
S1_ALT Projects - Google Sh
localhost:5173/safety-intelligence
Chat
UrbanRisk
INTELLIGENCE
Urban Traffic Risk Intelligence
AI-powered safety analytics
Prediction engine ready
AI
INTELLIGENCE
Overview
Risk Assessment
Risk Intelligence
Analytics
AI Explainability
PLATFORM
Safety Intelligence
RAG-powered safety intelligence
Safety Intelligence
Context-aware road-safety guidance grounded in the project's knowledge base and connected to the explainable accident-risk pipeline.
Knowledge layer active
RAG · 384-dim embeddings · grounded LLM
READY
Knowledge Base
Active
Road-safety guidance available for retrieval.
READY
Retrieval
12 chunks
indexed knowledge chunks in the local RAG store.
READY
AI Guidance
Grounded
LLM explanations use retrieved safety context.
Prediction and guidance are separate layers
The machine-learning model remains responsible for the accident risk classification. SHAP and LIME explain model behavior, while RAG retrieves safety knowledge and the LLM converts that context into user-facing guidance. Safety retrieval does not determine the predicted risk class.
Safety Guidance Library
Core domains represented in the project's road-safety knowledge base.
8 guidance domains
System Status
Online
XGBoost · SHAP · RAG · LLM
23°
ENG
US
13:49
29-10-2026


127.0.0.1:8000/health?utm
Urban Traffic Accident Risk
frontend
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UrbanRisk
INTELLIGENCE
Local feature-weight interpretation
Speed limit context
0.31
Road / junction context
0.24
Temporal context
0.19
Urban / rural context
0.16
Environmental context
0.10
Two complementary local explanation approaches
SHAP
PRIMARY
Feature attribution based on Shapley values.
LIME
Local surrogate model approximating the prediction.
SHAP and LIME explain model behavior from different methodological perspectives. Neither explanation should be interpreted as a causal analysis.
INTELLIGENCE
Overview
Risk Assessment
Risk Intelligence
Analytics
AI Explainability
PLATFORM
Safety Intelligence
Model Prediction
XGBoost determines the accident risk class.
SHAP Explanation
Shows feature contribution to model output.
LIME Explanation
Provides a local surrogate interpretation.
INTERPRETATION BOUNDARY
Explainability is not causality
SHAP and LIME describe how the trained machine learning model responds to the supplied features. They do not establish that a feature directly causes an accident outcome. The ML model remains responsible for risk classification.
System Status
Online
XGBoost · SHAP · RAG · LLM
23°
ENG
US
13:49
29-10-2026

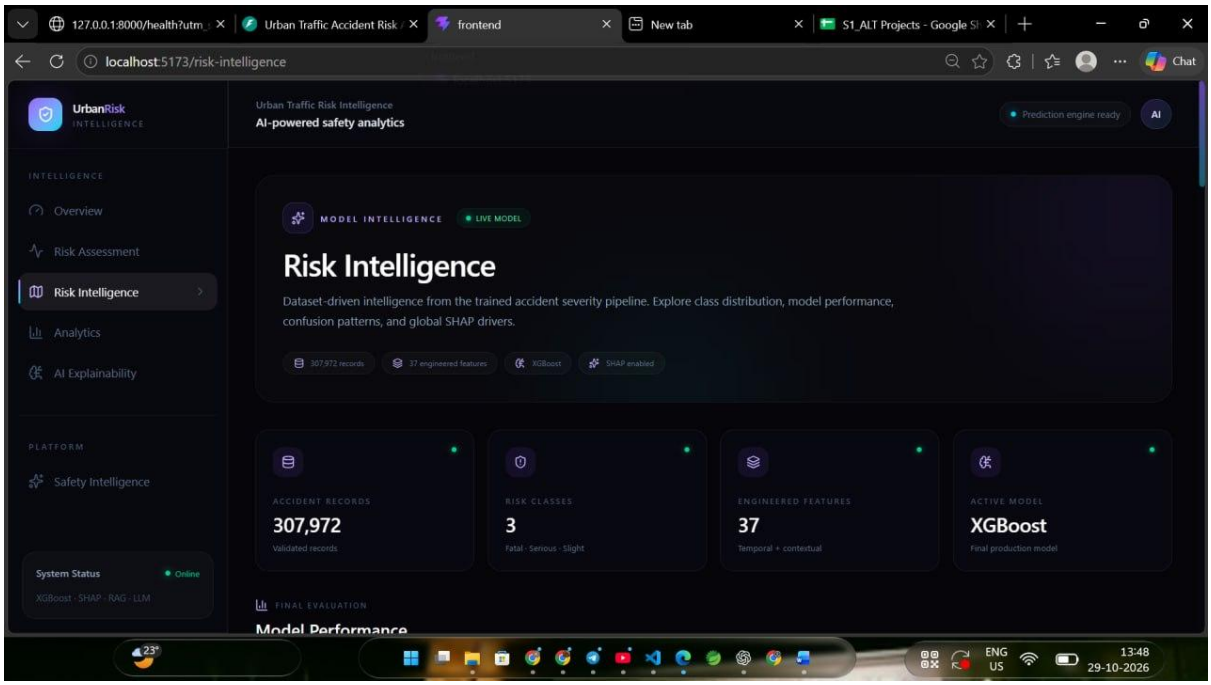

127.0.0.1:8000/health?utm
Urban Traffic Accident Risk
frontend
New tab
S1_ALT Projects - Google Sh
localhost:5173/explainability
Chat
UrbanRisk
INTELLIGENCE
Urban Traffic Risk Intelligence
AI-powered safety analytics
Prediction engine ready
AI
INTELLIGENCE
Overview
Risk Assessment
Risk Intelligence
Analytics
AI Explainability
PLATFORM
Safety Intelligence
EXPLAINABLE AI
SHAP ACTIVE
AI Explainability
Understand how the trained XGBoost model arrives at its predictions using global SHAP importance and local feature contributions, with LIME providing an additional local surrogate perspective.
SHAP
Local Explanation
LIME
3,000 Sample SHAP Analysis
Input
Accident context
XGBoost
Prediction
SHAP
Feature attribution
LIME
Local surrogate
Explanation
Human-readable
Prediction Snapshot
Example end-to-end model inference
PREDICTED RISK
Slight
MODEL
XGBoost
Input Context
Features supplied to the prediction pipeline
SPEED LIMIT
50 mph
ROAD TYPE
Single carriageway
URBAN / RURAL
Urban
WEATHER
Fine no high winds
ROAD SURFACE
Dry
LIGHTING
Daylight
System Status
Online
XGBoost - SHAP - RAG - LLM
23°
ENG
US
13:49
29-10-2026

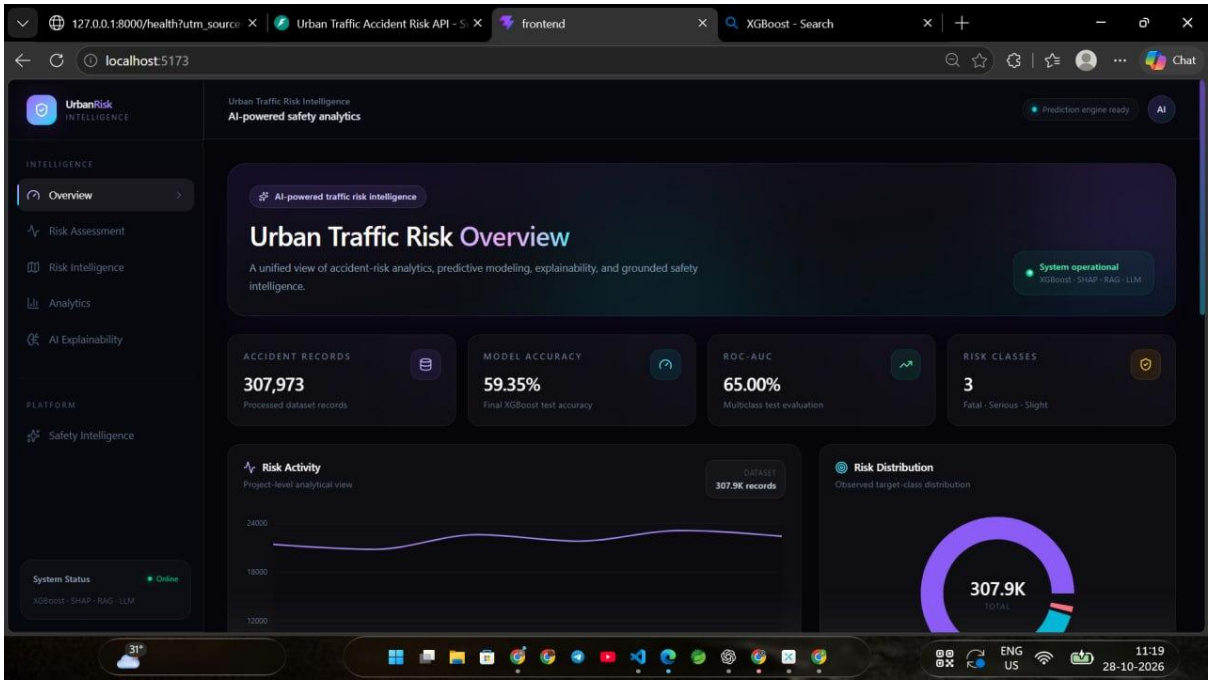

127.0.0.1:8000/health?utm
Urban Traffic Accident Risk
frontend
New tab
S1_ALT Projects - Google Sh
localhost:5173/risk-intelligence
Chat
UrbanRisk
INTELLIGENCE
Speed Limit Category
0.0298
INTELLIGENCE
Overview
Risk Assessment
Risk Intelligence
Analytics
AI Explainability
PLATFORM
Safety Intelligence
Confusion Matrix
Final XGBoost test-set predictions
ACTUAL \ PREDICTED
FATAL
SERIOUS
SLIGHT
Fatal
378
196
217
Serious
1,823
2,505
3,820
Slight
7,653
11,329
33,674
Reading the Matrix
Correct Fatal
378
Correct Serious
2,505
Correct Slight
33,674
The confusion matrix shows substantial cross-class confusion, particularly between Serious and Slight cases. This reflects the difficulty of separating minority severity classes in the imbalanced dataset.
INTELLIGENCE STACK
Predict → Explain → Ground → Assist
XGBoost performs the risk classification. SHAP explains model behavior. RAG retrieves relevant safety knowledge. The LLM converts these signals into grounded user-facing explanations.
XGBOOST
SHAP
RAG
LLM
System Status
Online
XGBoost - SHAP - RAG - LLM
23°
ENG
US
13:49
29-10-2026

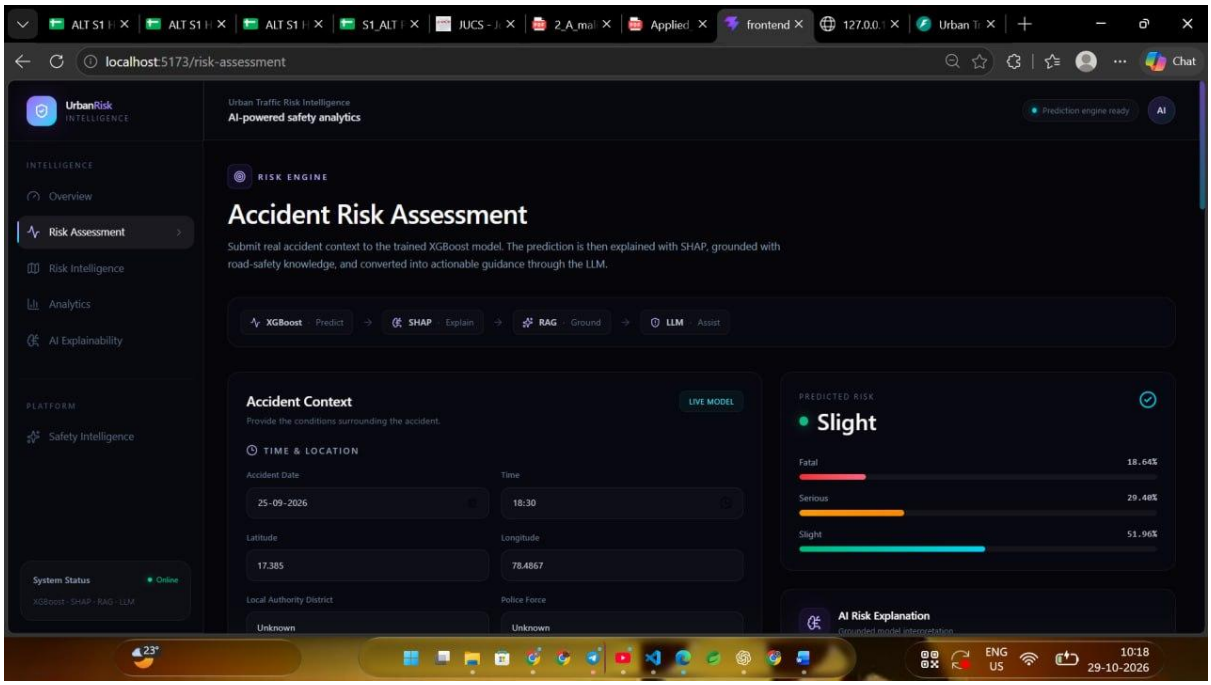

127.0.0.1:8000/health?utm
Urban Traffic Accident Risk
frontend
New tab
S1_ALT Projects - Google Si
localhost:5173/risk-intelligence
Chat
UrbanRisk
INTELLIGENCE
Speed Limit Category
0.0298
INTELLIGENCE
Overview
Risk Assessment
Risk Intelligence
Analytics
AI Explainability
PLATFORM
Safety Intelligence
Confusion Matrix
Final XGBoost test-set predictions
ACTUAL \ PREDICTED
FATAL
SERIOUS
SLIGHT
Fatal
378
196
217
Serious
1,823
2,505
3,820
Slight
7,653
11,329
33,674
Reading the Matrix
Correct Fatal
378
Correct Serious
2,505
Correct Slight
33,674
The confusion matrix shows substantial cross-class confusion, particularly between Serious and Slight cases. This reflects the difficulty of separating minority severity classes in the imbalanced dataset.
INTELLIGENCE STACK
Predict → Explain → Ground → Assist
XGBoost performs the risk classification. SHAP explains model behavior. RAG retrieves relevant safety knowledge. The LLM converts these signals into grounded user-facing explanations.
XGBOOST
SHAP
RAG
LLM
System Status
Online
XGBoost · SHAP · RAG · LLM
23°
ENG
US
13:49
29-10-2026


127.0.0.1:8000/health?utm
Urban Traffic Accident Risk
frontend
New tab
S1_ALT Projects - Google Si
localhost:5173/risk-intelligence
Chat
UrbanRisk
INTELLIGENCE
Model Comparison
Validation performance across the evaluated classifiers
70%
60%
40%
20%
0%
Logistic Regression
Decision Tree
Random Forest
XGBoost
Random Forest
Accuracy : 61.44%
Macro F1 : 34.87%
ROC-AUC : 63.77%
Accuracy
Macro F1
ROC-AUC
INTELLIGENCE
Overview
Risk Assessment
Risk Intelligence
Analytics
AI Explainability
PLATFORM
Safety Intelligence
MODEL
ACCURACY
MACRO F1
MACRO RECALL
ROC-AUC
Logistic Regression
41.71%
27.48%
43.86%
59.39%
Decision Tree
59.42%
34.12%
42.00%
52.93%
Random Forest
61.44%
34.87%
47.26%
63.77%
XGBoost
60.18%
35.12%
47.16%
65.33%
System Status
Online
XGBoost · SHAP · RAG · LLM
23°
ENG
US
13:48
29-10-2026


127.0.0.1:8000/health?utm
Urban Traffic Accident Risk
frontend
New tab
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localhost:5173/risk-intelligence
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UrbanRisk
INTELLIGENCE
INTELLIGENCE
Overview
Risk Assessment
Risk Intelligence
Analytics
AI Explainability
PLATFORM
Safety Intelligence
System Status
Online
XGBoost · SHAP · RAG · LLM
FINAL EVALUATION
Model Performance
Final held-out test-set performance of the deployed XGBoost classifier.
TEST
ACCURACY
59.35%
MACRO F1
34.74%
MACRO RECALL
47.49%
ROC-AUC
65.00%
Risk Class Distribution
Target distribution in the validated dataset
Production Model Snapshot
Final XGBoost test-set evaluation
ACCURACY
59.35%
Overall classification accuracy
MACRO PRECISION
37.00%
Class-balanced precision
MACRO RECALL
47.49%
Class-balanced recall
MACRO F1
34.74%
Balanced F1 across classes
WEIGHTED F1
ROC-AUC
23°
ENG
US
13:48
29-10-2026


127.0.0.1:8000/health?utm
Urban Traffic Accident Risk
frontend
New tab
S1_ALT Projects - Google Sh
localhost:5173/risk-intelligence
Chat
UrbanRisk
INTELLIGENCE
Urban Traffic Risk Intelligence
AI-powered safety analytics
Prediction engine ready
AI
INTELLIGENCE
Overview
Risk Assessment
Risk Intelligence
Analytics
AI Explainability
PLATFORM
Safety Intelligence
System Status
Online
XGBoost · SHAP · RAG · LLM
MODEL INTELLIGENCE
LIVE MODEL
Risk Intelligence
Dataset-driven intelligence from the trained accident severity pipeline. Explore class distribution, model performance, confusion patterns, and global SHAP drivers.
307,972 records
37 engineered features
XGBoost
SHAP enabled
ACCIDENT RECORDS
307,972
Validated records
RISK CLASSES
3
Fatal · Serious · Slight
ENGINEERED FEATURES
37
Temporal + contextual
ACTIVE MODEL
XGBoost
Final production model
FINAL EVALUATION
Model Performance
23°
ENG
US
13:48
29-10-2026


127.0.0.1:8000/health?utm_source
Urban Traffic Accident Risk API - S
frontend
XGBoost - Search
localhost:5173
Chat
UrbanRisk
INTELLIGENCE
Urban Traffic Risk Intelligence
AI-powered safety analytics
Prediction engine ready
AI
INTELLIGENCE
Overview
Risk Assessment
Risk Intelligence
Analytics
AI Explainability
PLATFORM
Safety Intelligence
AI-powered traffic risk intelligence
Urban Traffic Risk Overview
A unified view of accident-risk analytics, predictive modeling, explainability, and grounded safety intelligence.
System operational
XGBoost · SHAP · RAG · LLM
ACCIDENT RECORDS
307,973
Processed dataset records
MODEL ACCURACY
59.35%
Final XGBoost test accuracy
ROC-AUC
65.00%
Multiclass test evaluation
RISK CLASSES
3
Fatal · Serious · Slight
Risk Activity
Project-level analytical view
DATASET
307.9K records
24000
18000
12000
Risk Distribution
Observed target-class distribution
307.9K
TOTAL
System Status
Online
XGBoost · SHAP · RAG · LLM
31°
ENG
US
11:19
28-10-2026


ALT S1
S1_ALT
JUCS - J
2_A_mal
Applied
frontend
127.0.0.1
Urban Tr
localhost:5173/risk-assessment
Chat
UrbanRisk
INTELLIGENCE
Urban Traffic Risk Intelligence
AI-powered safety analytics
Prediction engine ready
AI
INTELLIGENCE
Overview
Risk Assessment
Risk Intelligence
Analytics
AI Explainability
PLATFORM
Safety Intelligence
RISK ENGINE
Accident Risk Assessment
Submit real accident context to the trained XGBoost model. The prediction is then explained with SHAP, grounded with road-safety knowledge, and converted into actionable guidance through the LLM.
XGBoost Predict
SHAP Explain
RAG Ground
LLM Assist
Accident Context
Provide the conditions surrounding the accident.
LIVE MODEL
TIME & LOCATION
Accident Date
25-09-2026
Time
18:30
Latitude
17.385
Longitude
78.4867
Local Authority District
Unknown
Police Force
Unknown
PREDICTED RISK
Slight
Fatal
18.64%
Serious
29.40%
Slight
51.96%
AI Risk Explanation
Grounded model interpretation
System Status
Online
XGBoost · SHAP · RAG · LLM
23°
ENG
US
10:18
29-10-2026